

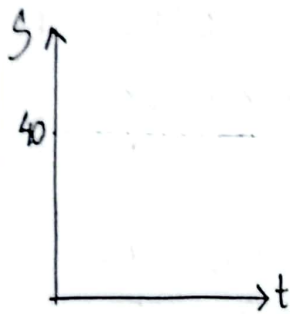
→ Calculus is basically the relation between two functions.

Function 1

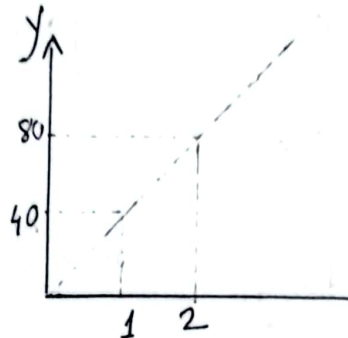
Distance
Height
→ $f(t)$ or $y(x)$

Function 2

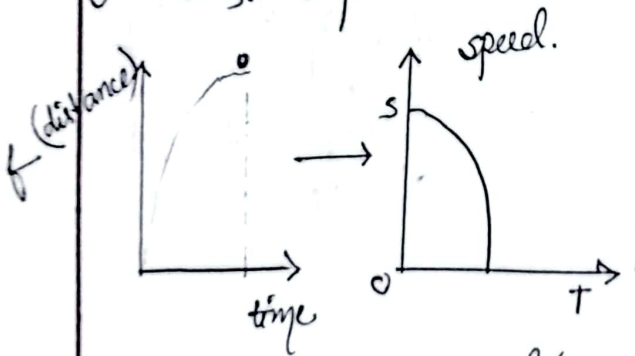
Speed → not constant
Slope
→ $s = \frac{df}{dt} / \frac{dy}{dx}$



graph of constant speed



$$\begin{aligned} \text{slope} &= \frac{\text{up}}{\text{across}} \\ &= \frac{40}{1} \\ &= \frac{80}{2} \\ &= 40. \end{aligned}$$



1. $f = st$

2. $s = b/t$

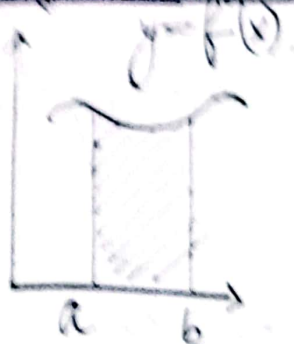
speed $\xrightarrow{\text{integral calculus}}$ distance

→ by finding the area of the curve

INTEGRATION

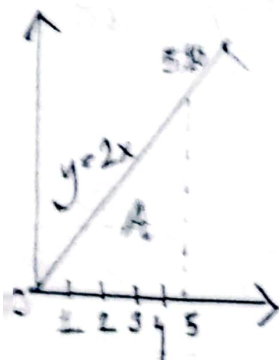
DIFFERENTIATION

Problem 1:



$$\text{area} = \int_a^b f(x) dx$$

start end



$$\begin{aligned} A &= \int_0^5 y \, dx \\ &= \int_0^5 2x \, dx \\ &= \left[x^2 \right]_0^5 \\ &= 25 \text{ unit}^2 \end{aligned}$$

Ans.

To compute the area:

1. We would divide the area into rectangles. They would be almost rectangles. (with smaller parts, curves become straight)

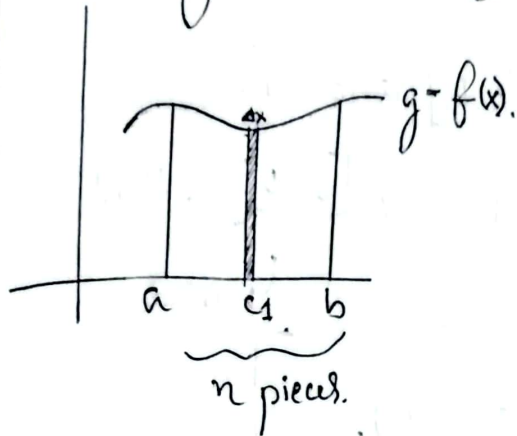
2. Calculate the area of individual parts and sum them up.

3. There would be some error calculating it. So, we would rectify it by taking as many rectangles as we can.

Lecture 02

* Riemann Sum:

General picture:



$$\Delta x = \frac{b-a}{n}$$

$$\sum_{i=1}^n \Delta x \cdot f(c_i) = \int_a^b f(x) \cdot dx.$$

(Riemann sum) (definite integral)

Ex:

t = time in years. (365 days).

$f(t)$ = \$/year borrowing rate

$$\Delta t = \frac{1}{365}$$

In day 45, $t = \frac{45}{365}$, your borrowed amount of 45th day.

$$f\left(\frac{45}{365}\right) \Delta t = f\left(\frac{45}{365}\right) \left(\frac{1}{365}\right).$$

$$\sum_{i=1}^{365} f\left(\frac{i}{365}\right) \Delta t = \int_0^1 f(t) dt.$$

↓
Riemann Sum notation.

* The fundamental theories of calculus: (FTC - I).

If $F'(x) = f(x)$, then $\int_a^b f(x) dx = f(b) - f(a)$

or,

If f is continuous on $[a, b]$ and f is any antiderivative of f on $[a, b]$, then
↳ closed.

$$\boxed{\int_a^b f(x) dx = f(b) - f(a).}$$

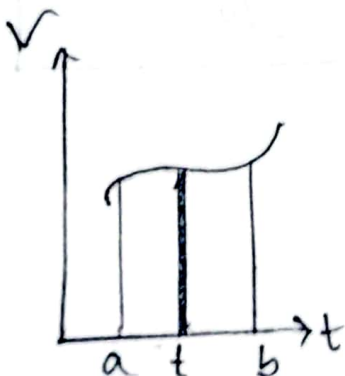
Ex: $f(x) = x^2 \quad \int_a^b x^2 dx = \left[\frac{x^3}{3} \right]_a^b = \frac{b^3}{3} - \frac{a^3}{3}.$

* Intuitive Interpretation of FTC - I:

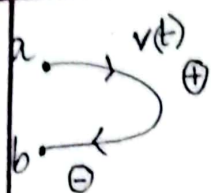
$x(t)$ = position at time t .

$x'(t) = \frac{dx}{dt}$ → speed

$$\left. \begin{array}{l} x(t) = \text{position at time } t. \\ x'(t) = \frac{dx}{dt} \rightarrow \text{speed} \end{array} \right\} \boxed{\int_a^b v(t) dt = x(b) - x(a)}$$



$$\begin{aligned} \sum_{i=1}^n v(t_i) \Delta t &= \int_a^b v(t) dt \\ &= x(b) - x(a). \end{aligned}$$



net distance $= \int_a^b v(t) dt$.

total distance $= \int_a^b |v(t)| dt$.

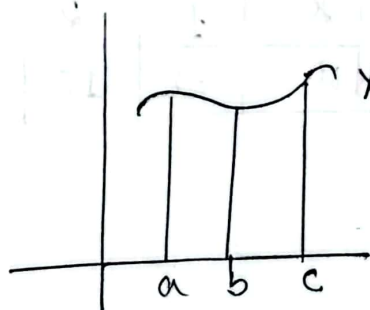
$$= \int_a^0 v(t) dt + \int_0^b v(t) dt.$$

* Properties of Definite Integral:

$$1. \int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx.$$

$$2. \int_a^b c f(x) dx = c \int_a^b f(x) dx.$$

3. For $a < b < c$



$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx.$$

$$4. \int_a^a f(x) dx = 0 \quad [F(a) - F(a) = 0]$$

$$5. \int_a^b f(x) dx = - \int_b^a f(x) dx.$$

$$F(b) - F(a) = -(F(a) - F(b))$$

[Estimation] 6. If $f(x) \leq g(x)$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$

7. Change of variables:

you have a function $u = u(x)$

$$\Rightarrow du = u'(x) dx.$$

$$\int_{u_1}^{u_2} g(u) du = \int g(u(x)) u'(x) dx$$

$$u_1 = u(x_1)$$

$$u_2 = u(x_2)$$

$$\longrightarrow \int_{u_1}^{u_2} g(u) du = \int_{x_1}^{x_2} g(u(x)) u'(x) dx$$

$$\int_1^2 (x^3 + 2)^5 x^2 dx.$$

$$u = x^3 + 2$$

$$du = 3x^2 dx.$$

$$x^2 dx = \frac{1}{3} du$$

x	1	2
u	3	10

$$\int_3^{10} u^5 \frac{1}{3} du.$$

$$= \frac{1}{3} \cdot \frac{1}{6} (10^6 - 3^6).$$

* FTC-1 (In reversed manner)

$$F(b) - F(a) = \int_a^b f(x) dx$$

$$\Delta F = F(b) - F(a) \quad \boxed{\Delta x = b - a}$$

$$\rightarrow \Delta F = \int_a^b f(x) dx$$

$$\Rightarrow \frac{\Delta F}{\Delta x} = \frac{1}{\Delta x} \int_a^b f(x) dx.$$

$$\frac{\Delta F}{\Delta x} = \frac{1}{b-a} \int_a^b f(x) dx$$

$$\Delta F = \left[\frac{1}{b-a} \int_a^b f(x) dx \right] \Delta x$$

$$\Delta F = [\text{Average of } F'] \Delta x.$$

$$\Delta F = F'(c) \Delta x$$

$$\min F' \leq F'(c) = \frac{1}{b-a} \int_a^b f(x) dx \leq \max F'$$

Example:

$f'(x) = \frac{1}{1+x}$, $F(0) = 1$ M.V.T. implies
that $A < F(4) < B$, what is A and B .

$$\frac{-5 \cos 5x}{-5 \sin 5x}$$

is

tr

According to MVT:

$$\Delta F = F'(c) \Delta x$$

$$\Rightarrow F(4) - F(0) = F'(c)(4 - 0) \quad [4 < c < 6]$$

$$\Rightarrow F(4) - 1 = \frac{1}{1+c} \cdot 4$$

$$\Rightarrow \max \left(\frac{1}{1+c} \cdot 4 \right) = \frac{1}{1+0} \cdot 4 [c=0] = 4$$

$$\min \left(\frac{1}{1+c} \cdot 4 \right) = \frac{1}{1+4} \cdot 4 [c=4] = 4/5.$$

$$\frac{4}{5} \leq F(4) - 1 \leq 4.$$

$$9/5 \leq F(4) \leq 5.$$

$$A = 9/5.$$

$$B = 5.$$

The fundamental theorem of calculus - 2:

If f is any continuous function and
 $G(x) = \int_a^x f(t) dt$ [$a \leq t \leq x$]
 then $G'(x) = f(x)$.

Example:

$\frac{d}{dx} \int_1^x \frac{dt}{t^2}$ $= \frac{d}{dx} \int_1^x \frac{1}{t^2} dt$ $= \frac{d}{dx} (G(x))$ $\therefore G'(x) = f(x) = \frac{1}{x^2}$	$f(t) = \frac{1}{t^2}$ $\therefore f(x) = \frac{1}{x^2}$ $G(x) = \int_1^x \frac{1}{t^2} dt$ $G(x) = \int_{a=1}^x f(t) dt.$
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$\Rightarrow$ Check FTC-01 and FTC-02:

1st part:

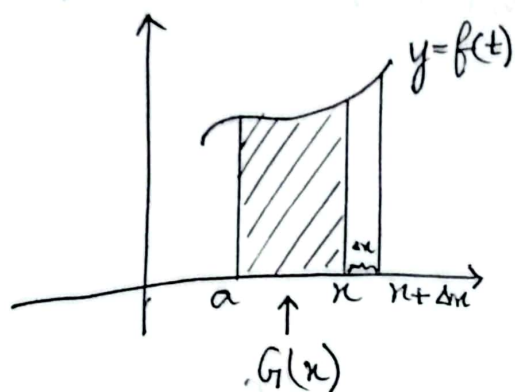
$$\int_1^x t^{-2} dt = [-t^{-1}]_1^x = -x^{-1} + 1 = -\frac{1}{x} + 1$$

(integration)

2nd part:

$$\frac{d}{dx} \left[\int_1^x \frac{dt}{t^2} \right] = \frac{d}{dx} \left(-\frac{1}{x} + 1 \right) = \frac{1}{x^2} \text{ (differentiation)}$$

Proof: (FTC-02)



$$\Delta G = \overset{\text{base}}{\Delta x} \cdot f(x).$$

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \frac{\Delta G}{\Delta x} = f(x).$$

$$\Rightarrow G'(x) = f(x)$$

$$G(x) = \int_a^x f(t) dt.$$

Proof: (FTC-01)

We started with $F'(x) = f(x)$ ——— (i)

From FTC-02, we have, $G(x) = \int_a^x f(t) dt.$

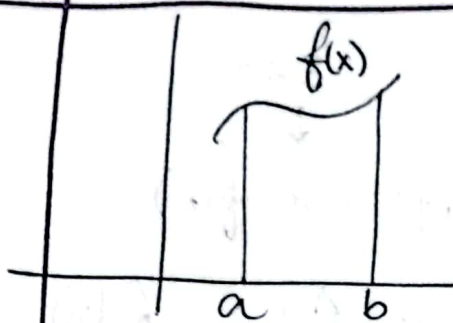
$$G'(x) = f(x) \text{ ——— (ii)}$$

From (i) and (ii), we get:

$$F'(x) = G'(x)$$

$$\Rightarrow F(x) = G(x) + C \text{ ——— [Integration on both sides]}$$

————— (iii)



$$F(x) = F(b) - F(a)$$

From (iii), we get:

$$\begin{aligned} F(b) - F(a) &= (G(b) + c) - (G(a) + c) \\ &= G(b) - G(a) \end{aligned}$$

$$\Rightarrow F(b) - F(a) = \int_a^b f(x) dx \quad [\text{FTC-02}]$$

Example:

$$L'(x) = \frac{1}{x}, \quad L(1) = 0$$

$$\text{FTC-02, } \boxed{L(x) = \int_1^x \frac{dt}{t}}$$

[Transcendental function]

$$\text{If } G(x) = \int_a^x f(t) dt.$$

$$G'(x) = f(x) = \frac{1}{x}.$$

$$\therefore f(x) = \frac{1}{x}$$

$$\therefore f(t) = \frac{1}{t}.$$

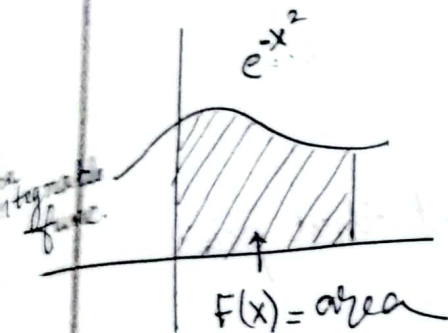
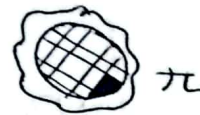
$$L(x) = \int_1^x \frac{1}{t} dt.$$

$$y' = e^{-x^2}$$

$$y(0) = 0$$

$$F(x) = \int_0^x e^{-t^2} dt \quad [\text{Transcendental function}]$$

↳ normal distribution



[Divide into rectangles
and solve by Riemann
Sum]

$$F(x) = \int_0^x e^{-t^2} dt$$

$$y = \frac{1}{x}$$

$$L(x) = \int_1^x \frac{dt}{t}$$

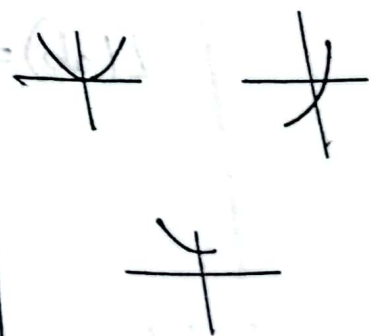
$$\Rightarrow L'(x) = \frac{1}{x}, \quad L(1) = \int_1^1 \frac{dt}{t} = 0$$

$$\Rightarrow L''(x) = -\frac{1}{x^2}$$

If second derivative of a function
is ~~has~~ negative, then curve of
that function will be concave
down.



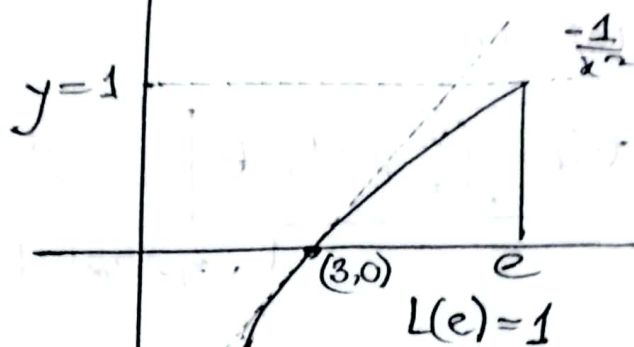
If positive,
concave up.



$$L'(x) = \frac{1}{x}, L'(1) = \frac{1}{1} = 1$$

$$L(1) = 0$$

$$L''(x) = -\frac{1}{x^2}$$



By FTC-02, we get:

1. Actual function.
2. Any point
3. Value of tangent at that point.
4. Second derivative.
5. The curve.

$$L(ab) = L(a) + L(b)$$

$$L(ab) = \int_1^{ab} \frac{dt}{t} = \int_1^a \frac{dt}{t} + \int_a^{ab} \frac{dt}{t} \quad \left. \begin{array}{l} L(ab) = L(b) \\ \text{to prove} \end{array} \right\} = L(a) + L(b)$$

Let, $t = au \Rightarrow u = t/a$
 $\therefore dt = a du$

$$L(b) = \int_a^{ab} \frac{dt}{t}$$

t	a	ab
u	1	b

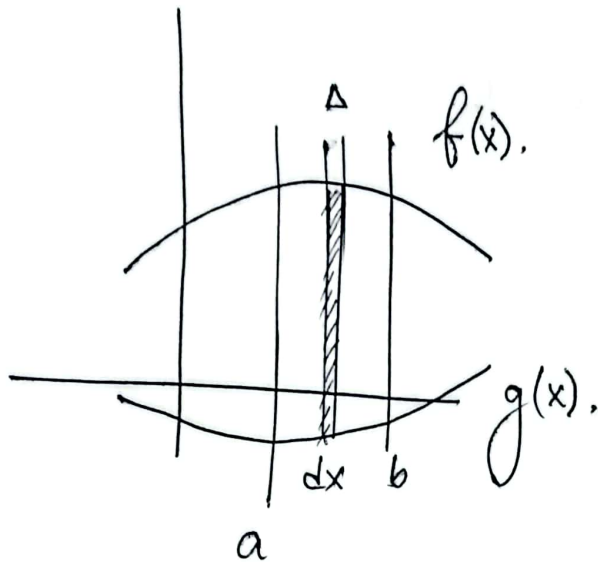
$$\int_a^b \frac{dt}{t} = \int_1^b \frac{a du}{a u} = \int_1^b \frac{du}{u} = L(b)$$

$$\therefore L(a) + L(b) = L(ab) = L(b)$$

MATH-4241: (L-4)

15/01/20

* Area between curves:



$$\Delta = \text{base} \cdot \text{height} \\ = dx \cdot (f(x) - g(x))$$

Area of this small rectangle (Riemann Sum)

Area of the white region

$$= \int_a^b f(x) - g(x) \cdot dx$$

limits

integrands.

Example-01:

two parameters.

Consider the function $y = f(x) = x$.
find the interval $[0, b]$.

* FTC-01:

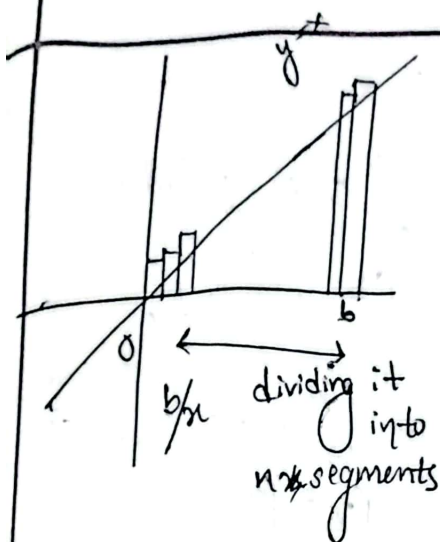
If $f(x) = F'(x)$,
then,

$$\int_a^b f(x) = F(b) - F(a)$$

* FTC-02:

If $G(x) = \int_a^x f(t) dt$,
then

$$G'(x) = f(x)$$



Let, n be a large positive integer and divide the interval $[0, b]$ into n equal ~~segments~~ subintervals.

$$x_1 = \frac{b}{n} \dots f(x_1) = \frac{b}{n}$$

$$x_2 = \frac{2b}{n} \dots f(x_2) = \frac{2b}{n}$$

$$x_3 = \frac{3b}{n} \dots f(x_3) = \frac{3b}{n}$$

$$x_{n-1} = \frac{(n-1)b}{n} \dots f(x_{n-1}) = \frac{(n-1)b}{n}$$

The whole area = sum of the area of all rectangles
[according to Riemann Sum]

$$= \left(\frac{b}{n}\right)\left(\frac{b}{n}\right) + \left(\frac{b}{n}\right)\left(\frac{2b}{n}\right) + \left(\frac{b}{n}\right)\left(\frac{3b}{n}\right) + \dots + \left(\frac{b}{n}\right)\left(\frac{nb}{n}\right)$$

$$= \frac{b^2}{n^2} [1 + 2 + 3 + \dots + n]$$

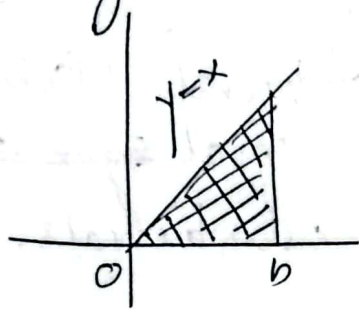
$$= \frac{b^2}{n^2} \times \frac{n(n+1)}{2}$$

$$= \frac{b^2}{2} \left(\frac{n+1}{n}\right) = \frac{b^2}{2} \left(1 + \frac{1}{n}\right)$$

if $\Rightarrow n \rightarrow \infty$
then, Area = $\frac{b^2}{2}$



According to Leibniz notation:

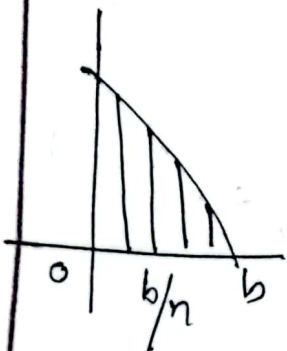


$$\begin{aligned}
 \text{Area} &= \int_0^b f(x) - g(x) dx \\
 &= \int_0^b (x - 0) dx \\
 &= \int_0^b x dx \\
 &= \left[\frac{x^2}{2} \right]_0^b \\
 &= \frac{b^2}{2}
 \end{aligned}$$

~~Example 01: Consider the function~~

~~$y = f(x) = x$, find~~

Example-01: Consider the function $y = \cos x$.
Find the interval $x=a$ to b .



$$\begin{aligned}
 x_1 &= \frac{b}{n} \dots f(x_1) = \cos \frac{b}{n} \\
 x_2 &= \frac{2b}{n} \dots f(x_2) = \cos \frac{2b}{n} \\
 &\vdots \\
 x_n &= \frac{nb}{n} \dots f(x_n) = \cos \frac{nb}{n}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area} &= \left(\frac{b}{n}\right) \left(\cos \frac{b}{n}\right) + \left(\frac{b}{n}\right) \left(\cos \frac{2b}{n}\right) + \dots + \left(\frac{b}{n}\right) \left(\cos \frac{nb}{n}\right) \\
 &= \frac{b}{n} \left[\cos \frac{b}{n} + \cos \frac{2b}{n} + \dots + \cos \frac{nb}{n} \right] \\
 &= \frac{b}{n} \sum_{k=1}^n \cos \frac{kb}{n} \quad \text{--- (i)} \quad \cos \text{hd} = \frac{\cos(n+1) \frac{b}{2n} \sin \frac{nb}{2n}}{\sin \frac{b}{2n}}
 \end{aligned}$$

$$\sum_{k=1}^n \left(\cos k \frac{b}{n} \right) = \frac{\cos(n+1) \frac{b}{2n} \sin \frac{nb}{2n}}{\sin \frac{b}{2n}}$$

Putting (ii) in (i),

$$= \frac{b}{n} \frac{\sin \frac{b}{2} \cos \left(1 + \frac{1}{n}\right) \frac{b}{2}}{\sin \frac{b}{2n}}$$

now, limit $\cos \left(1 + \frac{1}{n}\right) \frac{b}{2} = \cos \frac{b}{2}$
 $n \rightarrow \infty$

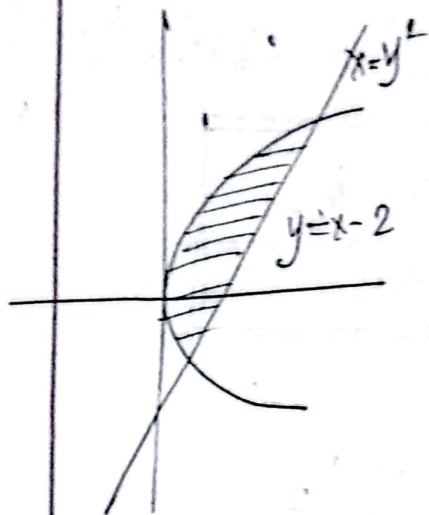
$$\frac{b}{n} \frac{1}{\sin \frac{b}{2n}} = 2 \frac{\frac{b}{2n}}{\sin \left(\frac{b}{2n}\right)} = 2$$

We get: $S_n = 2 \cdot \sin \frac{b}{2} \cdot \cos \frac{b}{2}$
 $= \sin b.$

Equivalently,
 $\int_0^b \cos x = \sin b.$

Example-03:

Find the area between $x=y^2$ and $y=x-2$



Vertical Slicing: (dx)

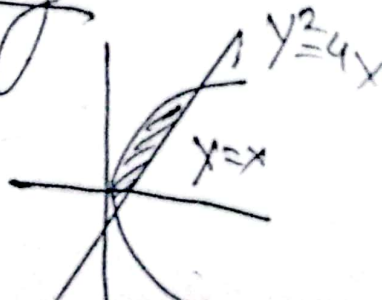
$$\begin{aligned} y &= y^{\pm 2} \\ \Rightarrow y^{\pm} - y - 2 &= 0 \\ \Rightarrow (y-2)(y+1) &= 0 \\ \Rightarrow y &= 2, y = -1 \\ \Rightarrow x &= 4, x = 1 \end{aligned}$$

$$\begin{aligned} \text{Area} &= \int_a^b \text{width} (f(x) - g(x)) dx \\ &= \int_0^1 \sqrt{x} - (-\sqrt{x}) + \int_1^4 \sqrt{x} - (x-2) \end{aligned}$$

=

Horizontal Slicing

$$y^2 = 4x$$



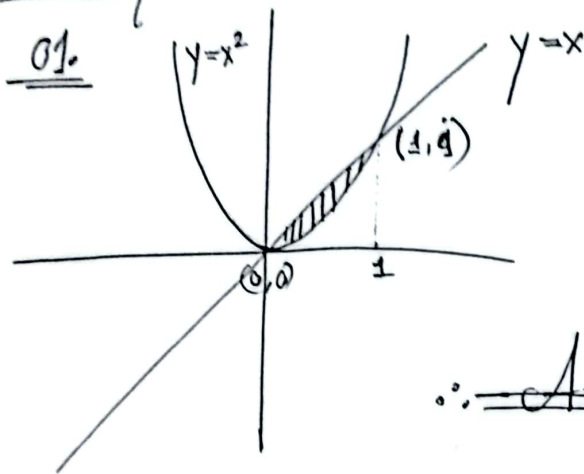
MATH-4241: (1-5)

17/01/23

Problem-01: Find by integration the area between curves $y=x$ and $y=x^2$ [with sketch]

Problem-02: Find by integration the area under curves $y^2=4x$ and $2x-y=4$ [with sketch]

Solution:



Given,

$$y_1 = x \quad \text{--- (i)}$$

$$y_2 = x^2 \quad \text{--- (ii)}$$

~~$\therefore$ Area~~

From (i) and (ii) $\Rightarrow$

$$x^2 = x$$

$$\Rightarrow x^2 - x = 0$$

$$\Rightarrow x(x-1) = 0$$

$$\therefore x = 0, 1.$$

$$\therefore \text{Area} = \int_0^1 (y_2 - y_1) dx$$

$$= \int_0^1 (x^2 - x) dx$$

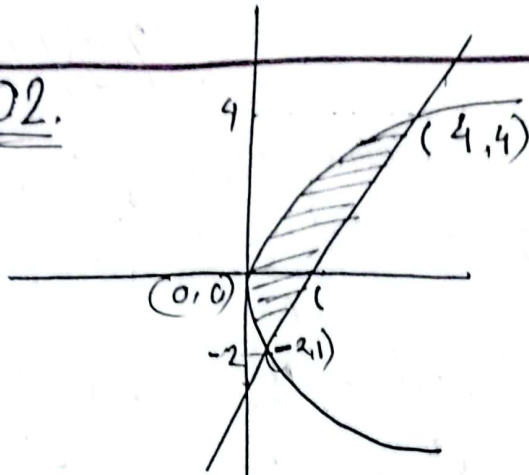
$$= \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_0^1$$

$$= \frac{1}{3} - \frac{1}{2}$$

$$= \frac{1}{6} \text{ (excluding the negative for area)}$$

sq. units

Q2.



Given,

$$y^2 = 4x, \text{--- (i)}$$
$$y = 2x - 4. \text{--- (ii)}$$

From (i) and (ii) $\Rightarrow$

$$y = 2\left(\frac{y^2}{4}\right) - 4$$
$$\Rightarrow y = \frac{y^2}{2} - 4$$
$$\Rightarrow y^2 - 2y - 8 = 0$$
$$\Rightarrow y = -2, 4.$$

$$\therefore \text{Area} = \int_{-2}^4 (x_2 - x_1) dy$$
$$= \int_{-2}^4 \left(\frac{y+4}{2} - \frac{y^2}{4} \right) dy.$$
$$= \int_{-2}^4 \left(\frac{y}{2} + 2 - \frac{y^2}{4} \right) dy$$
$$= \left[\frac{y^2}{4} + 2y - \frac{y^3}{12} \right]_{-2}^4$$
$$= \frac{20}{3} + \frac{4}{3}$$
$$= 8$$

Chapter-10s

* $\int x e^{-x^2} dx$

$\Rightarrow \int e^z \frac{dz}{2}$

$= -\frac{1}{2} e^z + C$

$= -\frac{1}{2} e^{-x^2} + C$

Let, $-x^2 = z$
 $\rightarrow -2x dx = dz$
 $\Rightarrow x dx = \frac{dz}{-2}$

Ans

* $\int \frac{dx}{\sqrt{9-4x^2}}$

$= \frac{1}{2} \int \frac{dx}{\sqrt{(\frac{3}{2})^2 - (x)^2}}$

$= \frac{1}{2} \cdot \frac{2}{3} \sin^{-1} \frac{x}{3/2} + C$

$= \frac{1}{3} \sin^{-1} \frac{2x}{3} + C$

Ans

* $\int \frac{x dx}{\sqrt{9-4x^2}}$

$= \int \frac{z dz}{-4z}$

$= -\frac{1}{4} z + C$

$= -\frac{1}{4} \sqrt{9-4x^2} + C$

Let, $9-4x^2 = z$
 $\Rightarrow -8x dx = 2 dz$

Ans

➔ Certain Trigonometric Integrals:

Ans

* $\int \sin^2 x \cos^2 x dx$

$= \frac{1}{4} \int (2 \sin x \cos x)^2 dx$

$= \frac{1}{4} \int \sin^2 2x dx$

$= \frac{1}{8} \int (1 - \cos 4x) dx$

$= \frac{1}{8} \left(x - \frac{\sin 4x}{4} \right) + C$

$= \frac{x}{8} - \frac{\sin 4x}{32} + C$

Ans

10.3:

$$1. \sqrt{a^2 - x^2} = 1 - \sin^2 \theta = \cos^2 \theta.$$

$$2. \sqrt{a^2 + x^2} = 1 + \tan^2 \theta = \sec^2 \theta.$$

$$3. \sqrt{x^2 - a^2} = \sec^2 \theta - 1 = \tan^2 \theta.$$

$$Q. \int \frac{\sqrt{x^2 - a^2}}{x} dx.$$

$$\text{Let, } x = a \sec \theta.$$

$$\therefore dx = a \sec \theta \tan \theta d\theta.$$

$$\begin{aligned} \sqrt{x^2 - a^2} &= \sqrt{(a \sec \theta)^2 - a^2} \\ &= a \tan \theta. \end{aligned}$$

$$\therefore \int \frac{a \tan \theta}{a \sec \theta} a \sec \theta \tan \theta d\theta$$

$$= \int a \tan^2 \theta d\theta.$$

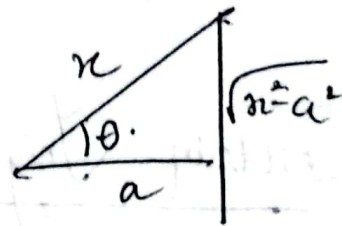
$$= a \int (\sec^2 \theta - 1) d\theta$$

$$= a \int \sec^2 \theta d\theta - \int d\theta$$

$$= a \tan \theta - a\theta + c$$

$$= a \frac{\sqrt{x^2 - a^2}}{a} - a \tan^{-1} \frac{\sqrt{x^2 - a^2}}{a} + c$$

$$= \sqrt{x^2 - a^2} - a \tan^{-1} \frac{\sqrt{x^2 - a^2}}{a} + c \quad \text{Any}$$



From figure:

$$\cos \theta = \frac{a}{x}$$

$$\Rightarrow \theta = \sec^{-1} \frac{x}{a}$$

$$\tan \theta = \frac{\sqrt{x^2 - a^2}}{a}$$

$$\begin{aligned}
 & \star 3 - (x^2 - 2x + 1) + 1 \rightarrow \text{Square Term} \\
 & = 4 - (x^2 - 2x + 1) \\
 & = 4 - (x-1)^2
 \end{aligned}$$

$$= 2^2 - (x-1)^2 \rightarrow \text{Completing the square.}$$

$$a^2 = x^2.$$

$$\int \frac{(x+2)dx}{\sqrt{3-2x-x^2}}$$

$$u = x-1.$$

$$a = 2.$$

$$x = u+1.$$

$$\Rightarrow dx = du.$$

$$x+2 = u+3.$$

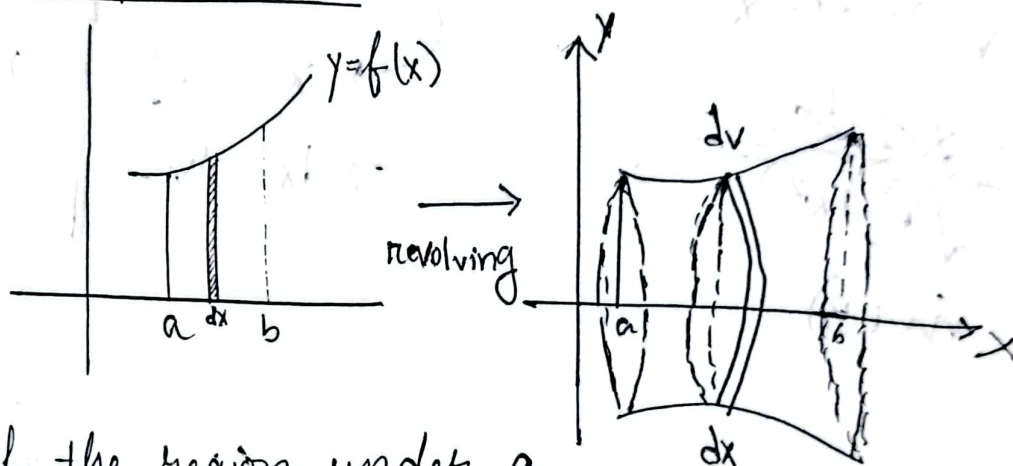
$$= \int \frac{(x+2)dx}{\sqrt{a^2 - x^2}}$$

$$= (u+3)$$

Ans.

Chapter-07:-(7.3) → Volume by Integration:-

1. Disk Method → Washer Method
2. Shell Method.

*DISK METHOD:

If the region under a curve $y=f(x)$ between $x=a$ and $x=b$ is revolved about X -axis, it generates a 3D shape. It's called solid of revolution.

The symmetrical shape of this solid makes its volume easy to compute.

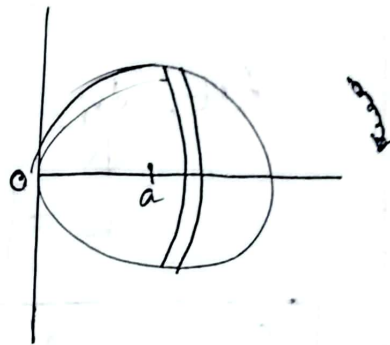
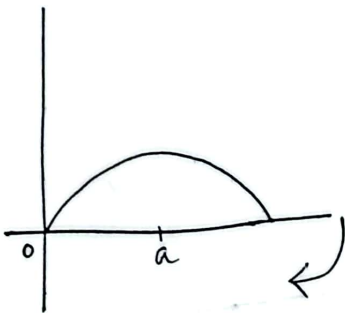
$$dV = 1 \cdot dx$$

$$= \pi y^2 dx$$

$$= \pi \cdot f(x)^2 \cdot dx$$

$$V = \int_a^b \pi f(x)^2 dx$$

2.



$$x^2 + y^2 = a^2$$

$$\Rightarrow y = \sqrt{a^2 - x^2}$$

$$dV = \pi y^2 dx$$

$$= \pi (a^2 - x^2) dx$$

$$V = 2 \int_0^a \pi (a^2 - x^2) dx$$

$$= 2 \left[\pi a^2 x - \frac{\pi x^3}{3} \right]_0^a$$

$$= 2 \left(\pi a^3 - \frac{\pi a^3}{3} \right)$$

$$= 2a^3 \frac{3\pi - \pi}{3}$$

$$= \frac{4}{3} \pi a^3 \text{ cub. unit. Ans.}$$

* Integration by parts:

If there are two types of differentials,

$$\frac{d}{dx}(uw) = u \frac{dw}{dx} + w \frac{du}{dx} \quad [\text{integrating both sides by } x]$$

$$\Rightarrow uw = \int u \frac{dw}{dx} dx + w \int \frac{du}{dx} dx.$$

$$\text{let, } \frac{dw}{dx} = v$$

$$w = \int v dx$$

$$\Rightarrow \int uv dx = u \int v dx - \int \left(\frac{du}{dx} \int v dx \right) dx.$$

$$* \int x^2 e^x dx$$

~~$$= e^x \int x^2 dx = \int \left(\frac{d}{dx} e^x \int x^2 dx \right) dx.$$~~

~~$$= \frac{e^x x^3}{3}$$~~

$$= x^2 \int e^x dx - \int \left(\frac{d}{dx} x^2 \int e^x dx \right) dx$$

$$= x^2 e^x - \int 2x e^x dx$$

$$= x^2 e^x - 2 \left[x \int e^x dx - \int \left(\frac{dx}{dx} \int e^x dx \right) dx \right]$$

$$= x^2 e^x - 2x e^x + 2 \int e^x dx$$

$$= x^2 \cdot e^x - 2x \cdot e^x + 2 \cdot e^x$$

* The method of partial fractions:

proper = numerator < denominator (degree)
improper = numerator > denominator (")

$$\frac{x^2+2}{x(x^2-9)} \rightarrow \text{proper}$$

$$\frac{2x^3-3x^2+4}{x^2+4} \rightarrow \text{improper.}$$

* proper:

$$\int \frac{6x^2+14x-20}{x^3-4x} dx$$

$$\Rightarrow \int \frac{6x^2+14x-20}{x(x-2)(x+2)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x-2}$$

$$\Rightarrow 6x^2+14x-20 = A(x-2)(x+2) + Bx(x-2) + Cx(x+2).$$

$$\text{If } x=0 \rightarrow -20 = -4A \\ \Rightarrow A=5.$$

$$\text{If } x=2 \rightarrow 32 = 8C \\ \Rightarrow C=4$$

$$\text{If } x=-2 \rightarrow -24 = 8B. \\ \Rightarrow B=-3$$

$$\therefore \int \frac{6x^2 + 14x - 20}{x(x-2)(x+2)} dx$$

$$= \int \left(\frac{5}{x} - \frac{3}{x+2} + \frac{4}{x-2} \right) dx.$$

$$= 5 \ln x - 3 \ln |x+2| + 4 \ln |x-2| + c$$

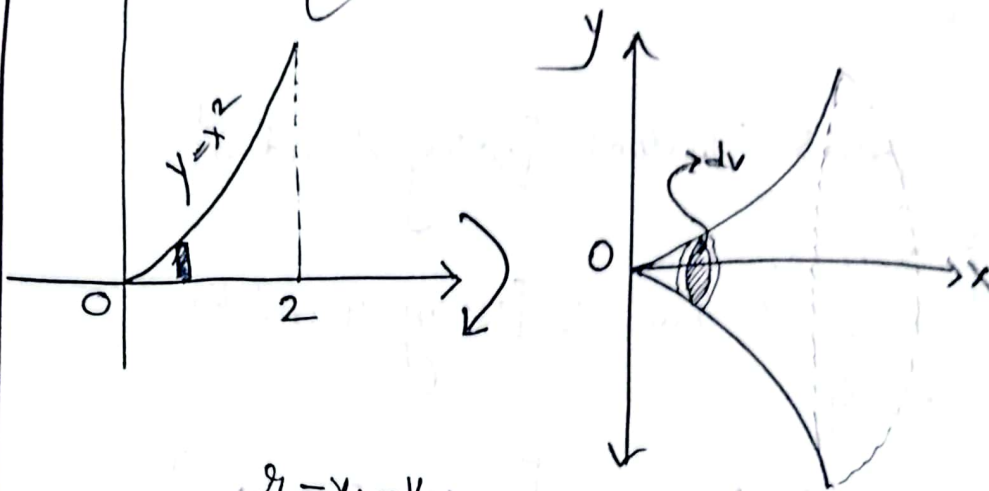
Ans.

* improper:

$$\frac{x^3 + 3x^2 + 2}{x+2}$$

$$\frac{f(x)}{c(x)} = \text{polynomial} + \frac{f'(x)}{c(x)}$$

Disk Method around X-axis:

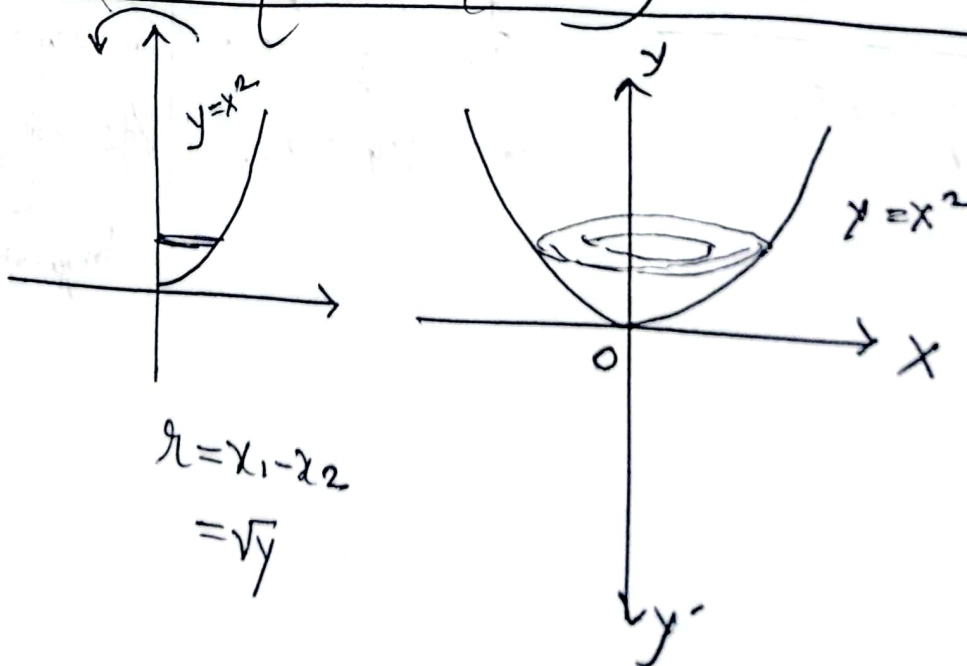


$$\begin{aligned}
 r &= y_1 - y_2 \\
 &= x^2 - 0 \\
 &= x^2
 \end{aligned}$$

$$\begin{aligned}
 dV &= A \cdot dx \\
 &= \pi (x^2)^2 dx \\
 &= \pi x^4 dx
 \end{aligned}$$

$$V = \int_0^2 \pi x^4 dx$$

Disk Method around y-axis:



$$\begin{aligned}
 r &= x_1 - x_2 \\
 &= \sqrt{y}
 \end{aligned}$$

$$dV = A dx$$

$$= \pi y dx$$

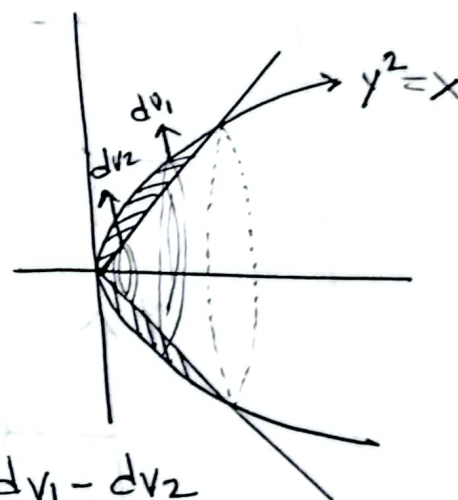
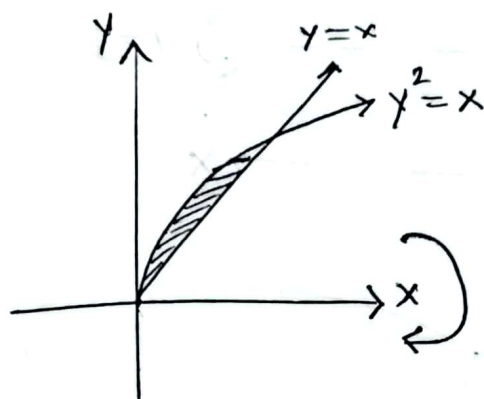
$$V = \int_a^b \pi y dy$$

$$A = \pi (G(y))^2$$

$$= \pi (\sqrt{y})^2$$

$$= \pi y$$

Solid of revolution betⁿ 2 functions:



$$V = dV_1 - dV_2$$

$$\sqrt{x} = x$$

$$\Rightarrow x^2 - x = 0$$

$$\Rightarrow x(x-1) = 0$$

$$\therefore x = 0, 1.$$

$$dV_1 = dV_1$$

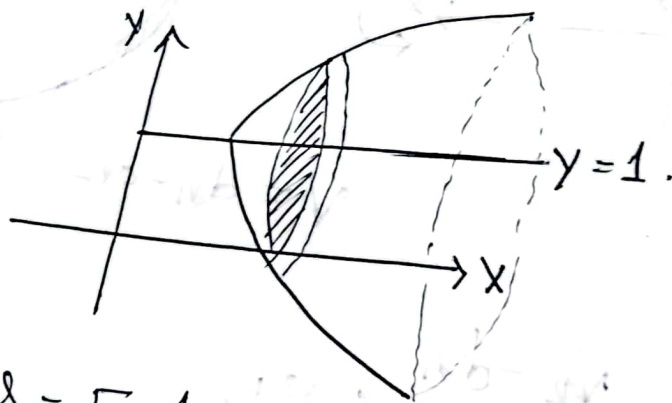
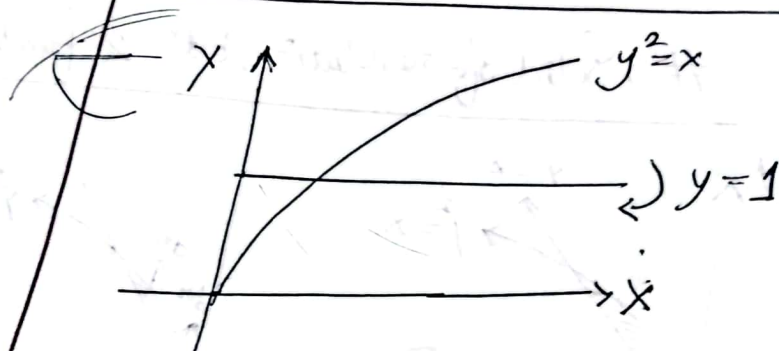
$$= \int_0^1 \pi (\sqrt{x})^2 dx$$

$$dV_2 = dV_2$$

$$= \int_0^1 \pi (x)^2 dx$$



Disk method around any horizontal line
(not x-axis):

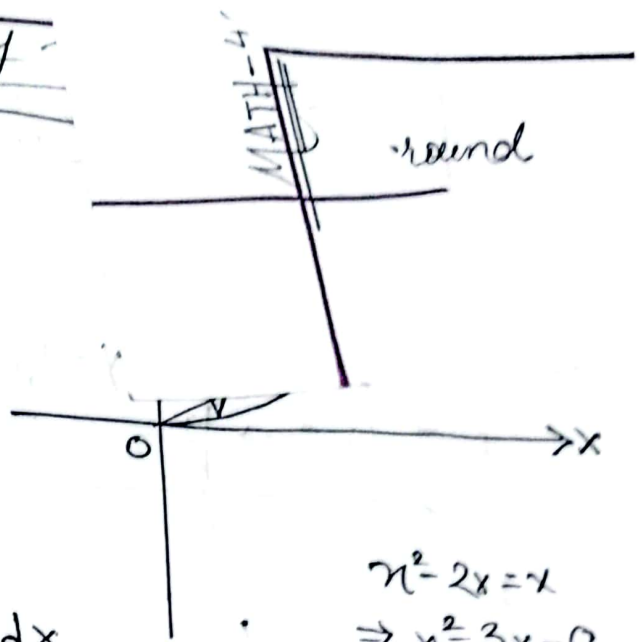
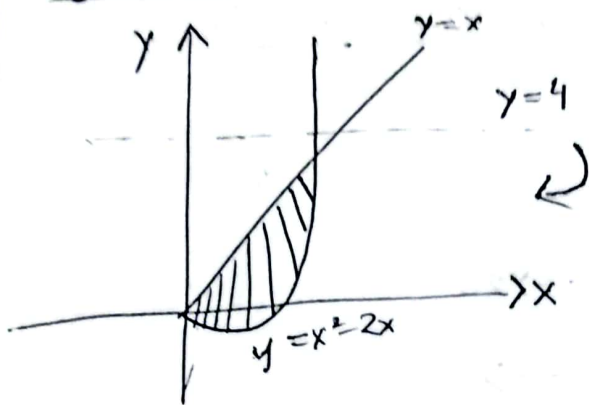


$$\text{radius} = \sqrt{x} - 1.$$

$$V = \int_a^b \pi r^2 dx.$$

$$= \int_a^b \pi (\sqrt{x} - 1)^2 dx.$$

Washer Method around any



$$V = \int_0^3 \pi (x^2 - 2x - 4)^2 dx$$

$$= \int_0^3 \pi (x-4)^2 dx$$

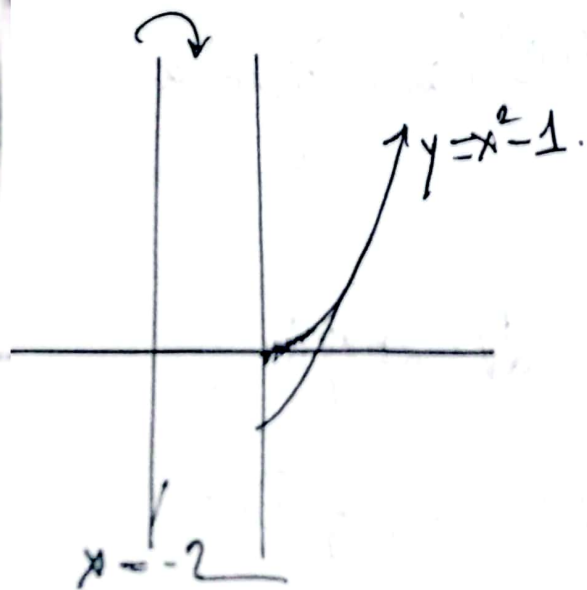
$$= \pi \int_0^3 (x^2 - 8x + 16) dx$$

$$= \pi \left[\frac{x^3}{3} - \frac{8x^2}{2} + 16x \right]_0^3$$

$$= 21\pi \text{ sq. units}$$

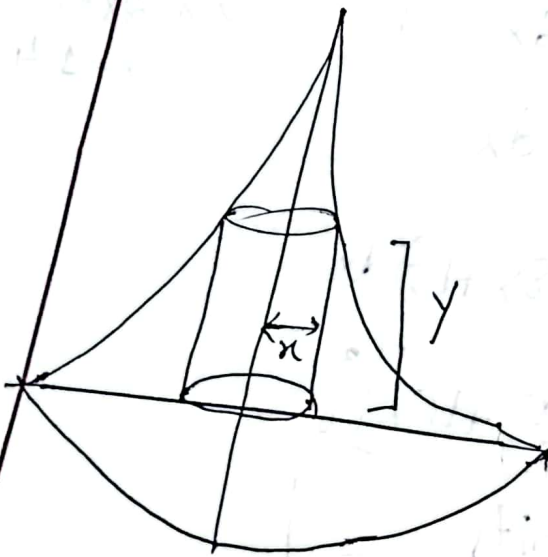
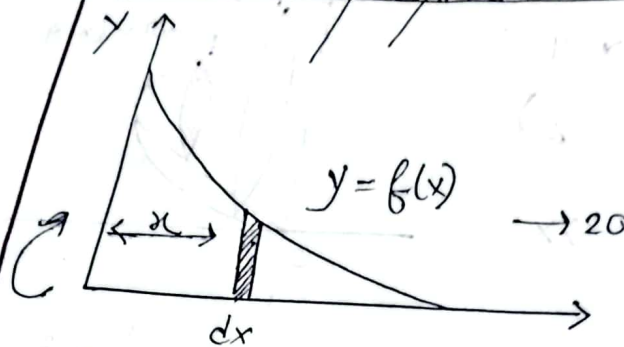
Ans

$$\begin{aligned} x^2 - 2x &= x \\ \Rightarrow x^2 - 3x &= 0 \\ \therefore x &= 1, 4 \end{aligned}$$



#3

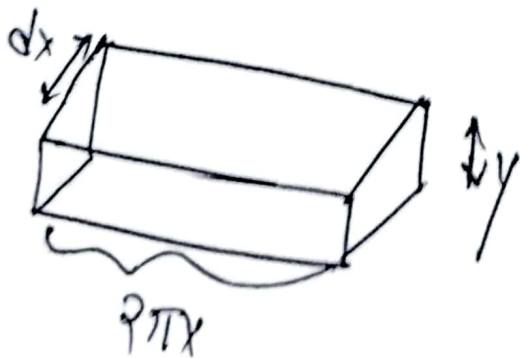
Volume by Cylindrical Shell Method:



$$h = f(x) - g(x)$$

$$\begin{aligned} \text{height} &= y \\ \text{radius} &= x \end{aligned}$$

$$V = \int_a^b 2\pi (\text{shell radius}) (\text{shell height}) dx$$



$$dV = 2\pi x y dx$$

$$V = \int 2\pi x f(x) dx$$

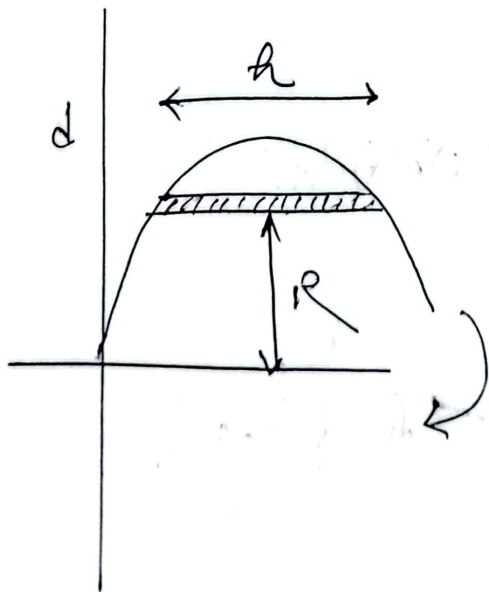
Disk vs. Shell:

* Disk method is suitable for restrictions around x -axis and shell method is suitable for y -axis.

* Disk/washer method, slice is perpendicular to the axis of revolution, in shell, it's parallel to the axis of revolution.

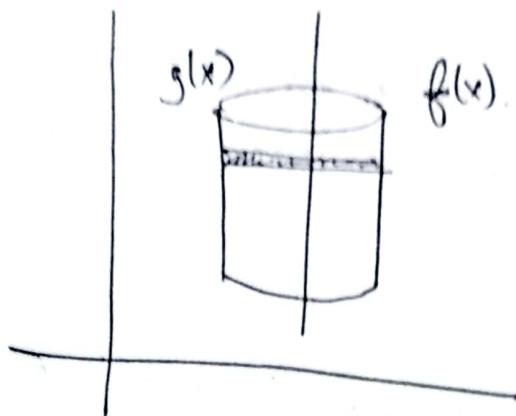
* In disk, taking disk as cross-section.
In shell, " shell " " " "

x -Axis:



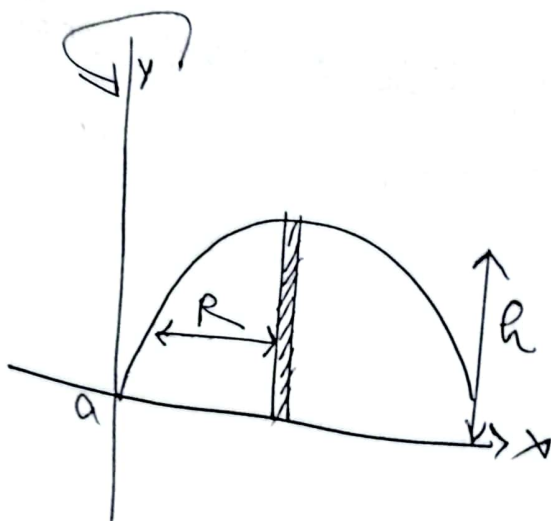
slice parallel

$$V = 2\pi \int_0^d R(y)h(y) dy.$$

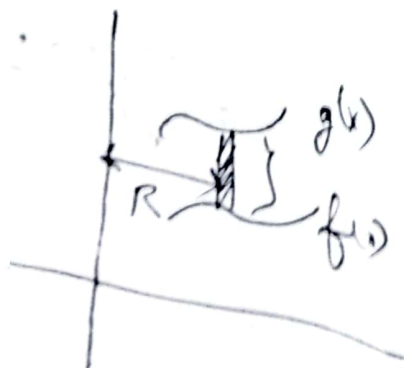


Y-axis:

$$V = 2\pi \int R(y) h(f(y) - g(y)) dy$$



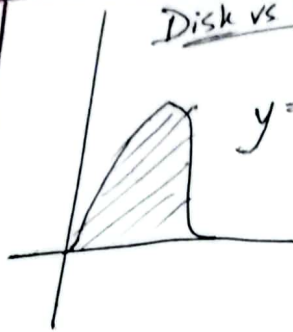
$$V = 2\pi \int_a^b R(x) h(x) dx$$



$$h = f(x) - g(x)$$

$$P \rightarrow q = P \cdot q$$

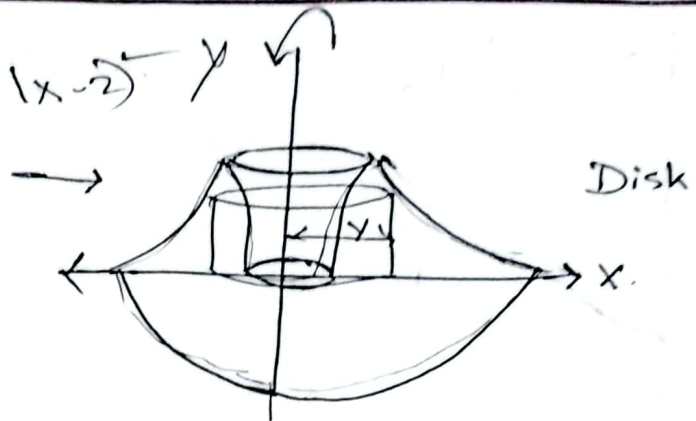
Disk vs Shell



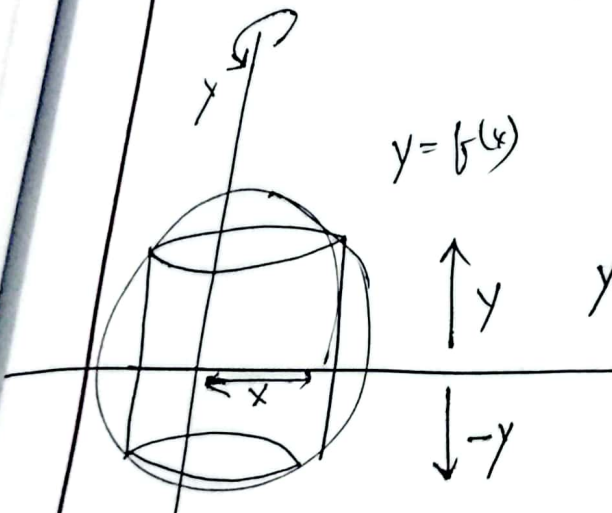
$$y = (x-1)(x-2)$$

$$h =$$

$$h =$$



$$V = \int_1^3 2\pi x (x-2)^2 (x-1) dx$$



$$y = f(x)$$

$$y^2 = a^2 + x^2$$

$$h = f(x) - g(x)$$

$$= \sqrt{a^2 + x^2} - (-\sqrt{a^2 + x^2})$$

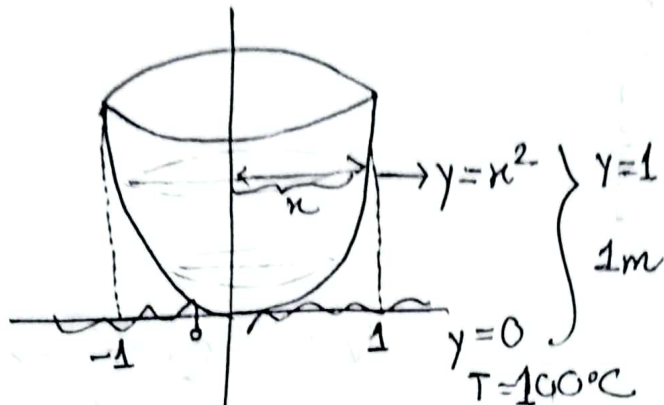
$$= 2\sqrt{a^2 + x^2}$$

$$= 2y$$

$$V = \int_0^a 2\pi x (2y) dx$$

$$= \int_0^a 4\pi x \sqrt{a^2 + x^2} dx$$

- * Weighted Average.
- * Arc length.
- * Miscellaneous Problem.



Initial Temperature = 0°C .

Final Temperature = T
 $= (100 - 30y)$

limit $[-1, 1]$

$$\therefore T = 100 - 30 = 70^\circ\text{C}$$

Q. How much energy we need to ~~reach~~ reach the final scenario?

$$\begin{aligned} \text{Energy} &= \text{Volume} \times \text{Temperature} \\ &= \int_0^1 v \cdot T \end{aligned}$$

$$= \int_0^1 T \cdot \pi x^2 dx$$

$$= \int_0^1 T \pi x^2 dy = \int_0^1 (100 - 30y) \cdot y dy = 45 \pi$$

Calories

1 Calorie = The amount of energy needed to increase 1°C temp. of 1 gm water.

$$1\text{cm}^3 - 1^{\circ}\text{C} \rightarrow 1\text{cal}$$

$$10\text{cm}^3 - 1^{\circ}\text{C} - 10\text{ cal.}$$

$$\begin{aligned}\text{Average Temperature} &= \frac{T_{\max} + T_{\min}}{2} \\ &= \frac{100 + 70}{2} \quad \times \\ &= 85^{\circ}\text{C} \quad \text{Ans.}\end{aligned}$$

$$\text{Real average Temperature} = \frac{\int f(x)w(x)dx}{\int w(x)dx}$$

$$\frac{\int_0^1 T y dy}{\int_0^1 H(y) dy} = \frac{V \cdot T}{V}$$

* Weighted Average: It's an average in which some of the items to be managed are 'more imp' or 'less imp' than the others. The weights are numbers which measure the relative importance.

$$\text{Formula} = \frac{\int_a^b f(x)w(x)dx}{\int_a^b w(x)dx}$$

Average of a list of numbers.

$$x_1, x_2, \dots, x_n$$

$$\frac{x_1 + x_2 + \dots + x_n}{n}$$

Average of f on interval $[a, b]$

$$\frac{1}{b-a} \int_a^b f(x) dx$$

$$y = x^2 \quad [2, 4] = \int_2^4 x^2 dx$$

$$= \left[\frac{x^3}{3} \right]_2^4 \times \frac{1}{2}$$

$$= \frac{1}{6} (4^3 - 2^3)$$

$$= 9.33$$

Ans.

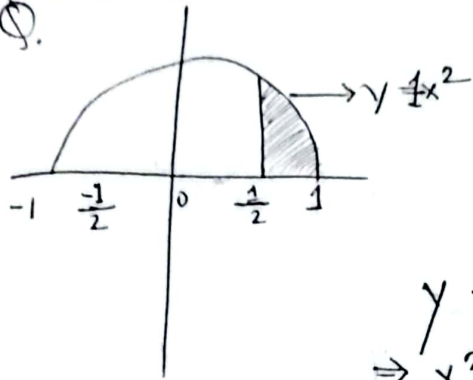
$$w_1 x_1 + w_2 x_2 + \dots + w_n x_n$$

$$w_1 + w_2 + \dots + w_n$$

Weighted average of a list of numbers.

$$\frac{\int_2^4 x^2 \cdot x dx}{\int_2^4 x dx}$$

Q.



Pick a point at random $0 < y < 1 - x^2$
and find the chances
of $|x| > 1/2$?

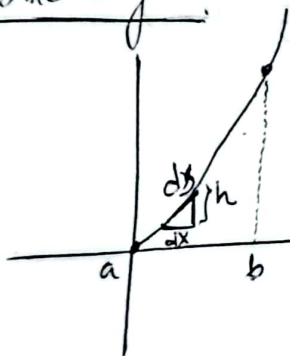
$$y = 1 - x^2$$

$$\Rightarrow x^2 = 1$$

$$\Rightarrow x = \pm 1.$$

$$\text{prob} = \frac{\int_{1/2}^1 (1 - x^2) dx}{\int_{-1}^1 (1 - x^2) dx} = \frac{\text{one part}}{\text{whole}}$$

* Arc length:



$$y = f(x).$$

$$ds^2 = dx^2 + dy^2$$

$$\Rightarrow ds = \sqrt{dx^2 + dy^2}$$

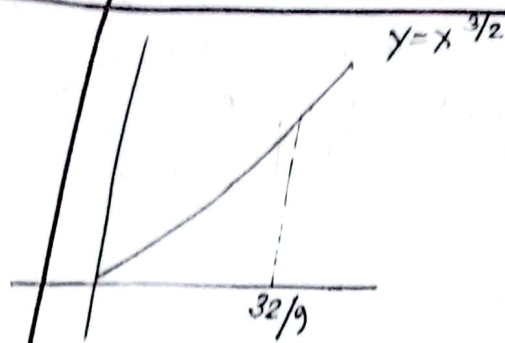
$$\int_a^b ds = \int_a^b \sqrt{dx^2 + dy^2}$$

$$= \int_a^b \sqrt{dx^2 \left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}}$$

$$= \int_a^b \sqrt{1 + \frac{dy^2}{dx^2}} dx$$

$$ds = \int_a^b \sqrt{1 + \{f'(x)\}^2} dx$$





Arc length

$$= \int_a^b \sqrt{1 + \{f'(x)\}^2} dx$$

$$= \int_0^{\frac{32}{9}} \sqrt{1 + \frac{9x}{4}} dx$$

$$= \frac{4}{9} \int_1^9 \sqrt{u} du$$

$$= \frac{4}{9} \left[\frac{2}{3} u^{3/2} \right]_1^9$$

$$f(x) = x^{3/2}$$

$$f'(x) = \frac{3}{2} x^{1/2}$$

$$\{f'(x)\}^2 = \frac{9}{4} x$$

$$u = 1 + \frac{9x}{4}$$

$$\Rightarrow du = \frac{9}{4} dx$$

$$\Rightarrow dx = \frac{4}{9} du$$

$$\left[\begin{array}{l} x=0 \rightarrow u=1 \\ x=\frac{32}{9}, u=9 \end{array} \right]$$

6.5

Ans